

Highlights

CET-STGL: A Causal-Inspired Event-Token Spatio-Temporal Framework for Alarm-Aware KPI Degradation Prediction in Cellular Networks

Dun Li, RuoShi Liu, Ming Li, Ruiguan Lin, Enrico Zio

- Private cellular KPI/alarm telemetry is studied under leakage-controlled forecasting.
- CET-STGL organizes KPI patches, sparse alarms, cell/time context and inferred cross-cell context.
- Forecasting, weak degradation classification and weak alarm-candidate prioritization are evaluated.
- Reported evidence is limited to runnable lightweight instantiations and weak labels.

CET-STGL: A Causal-Inspired Event-Token Spatio-Temporal Framework for Alarm-Aware KPI Degradation Prediction in Cellular Networks

Dun Li^{a,*}, Ruoshi Liu^a, Ming Li^b, Ruiguan Lin^a, Enrico Zio^{c,d}

^a*Department of Industrial Engineering, Tsinghua University, Beijing, 100084, China*

^b*Research Institute for Advanced Manufacturing, The Hong Kong Polytechnic University, Hung Hom, Hong Kong SAR, China*

^c*Center for research on Risk and Crises (CRC), Mines Paris, PSL, France*

^d*Energy Department, Politecnico di Milano, Milano, Italy*

Abstract

Cellular network reliability is monitored through heterogeneous time-evolving telemetry, including key performance indicator (KPI) measurements, business counters, sparse alarm/fault events, anonymized cell identifiers and temporal context. Predicting future KPI degradation is difficult because alarms are sparse, KPI responses may be delayed, physical topology is often unavailable and expert-confirmed fault labels are not always recorded. We study an anonymized real-world mobile broadband (MBB) operations dataset provided by Huawei Technologies. The source collection covers approximately 500 cellular sites, while the aligned panel evaluated in this study contains 33 anonymized cell entities without geographically identifiable site information. We propose CET-STGL, a causal-inspired event-token spatio-temporal framework that combines KPI patches, alarm events, cell/time context, train-only event influence priors and inferred cross-cell context. A lightweight sklearn-based instantiation is evaluated under a leakage-controlled protocol for multi-horizon KPI forecasting, weak degradation classification and weak

*Corresponding author.

Email addresses: lidun.2023@tsinghua.org.cn (Dun Li), liurs23@mails.tsinghua.edu.cn (Ruoshi Liu), ming.li@polyu.edu.hk (Ming Li), linruiguan@mail.tsinghua.edu.cn (Ruiguan Lin), enrico.zio@polimi.it (Enrico Zio)

URL: <https://orcid.org/0000-0003-4119-7340> (Ming Li), <https://orcid.org/0000-0003-2035-3770> (Ruiguan Lin)

alarm-candidate prioritization. Results indicate that KPI-history baselines remain strong, standalone event-only predictors are weak, and the evaluated CET-STGL instantiation is competitive on several primary-KPI forecasting metrics. Alarm and graph features show limited and non-uniform aggregate forecasting gains; weak alarm-candidate ranking is interpreted only as proxy consistency, not expert-confirmed fault localization.

Keywords: Cellular network reliability, KPI forecasting, Alarm correlation, Weak alarm-candidate prioritization, Spatio-temporal framework, Causal-inspired event influence

1. Introduction

Cellular network reliability is increasingly managed through structured operational telemetry rather than isolated counters [1, 2]. A cell-level operations panel may contain continuous key performance indicator (KPI) and business counters, sparse alarm/fault events, anonymized cell identifiers, and time context [3, 4]. KPI histories describe load, accessibility, retainability, throughput and radio-quality dynamics; alarms mark discrete operational events; cell and time features encode persistent local behavior and periodic demand. Anticipatory mobile networking has shown that prediction can support proactive resource management [5]. Recent 5G-Advanced and O-RAN perspectives place AI/ML closer to radio access automation, while O-RAN tutorials describe programmable interfaces and deployment choices for network management [6]. Recent O-RAN work stresses explainable automation [7]. Causal-reasoning discussions in AI-native wireless networks also motivate cautious treatment of observational associations [8]. For cellular operations, proactive KPI degradation prediction is a reliability problem: the goal is to recognize deteriorating cell conditions before they become service-impacting events or costly manual troubleshooting cases.

The modeling difficulty lies in the heterogeneous form and uneven reliability of the available signals. KPI-history models can exploit strong temporal regularities, yet sparse event context is often part of the evidence that operators inspect during fault handling. Alarm/event models alone are also insufficient for continuous KPI prediction because alarms are rare and many KPI changes follow routine traffic or radio patterns rather than explicit faults. Prior cellular traffic work has emphasized spatial and temporal dependencies among base stations [9], while recent mobile traffic forecasting stud-

ies show the importance of network-wide temporal patterns [10]. Classic alarm-correlation work has long noted that a single fault may generate many related alarms [11]; communication-network fault-identification studies also show that alarm floods complicate localization [12]. Enterprise operations data add further constraints: physical topology may be unavailable, expert root-cause annotations may be absent, and data release may be restricted by business and network-operation confidentiality.

This paper studies alarm-aware KPI degradation prediction using an anonymized real-world mobile broadband (MBB) operations dataset provided by Huawei Technologies. The source collection covers approximately 500 cellular sites, while the aligned panel evaluated in this study contains 33 anonymized cell entities without geographically identifiable site information. We propose CET-STGL as a causal-inspired event-token spatio-temporal framework for this setting. The central methodological idea is a temporally admissible representation of heterogeneous cellular telemetry: KPI histories are transformed into local patch features, alarms are represented through sparse event and intensity features, and alarm-degradation associations are incorporated as train-only event influence priors. When physical topology is unavailable, cross-cell context is inferred from training-period operational correlations. CET-STGL organizes operational signals under realistic monitoring constraints, without treating observed associations as causal evidence or expert-verified fault localization. The reliability-oriented pipeline and the main guardrails used throughout the paper are summarized in Fig. 1.

Recent telecom LLM discussions examine language models for telecommunications [13]. Domain-adapted telecom LLMs focus on language and standards knowledge [14], and networking LLM systems study adaptation to networking tasks [15]. Benchmark proposals for industrial telecom workflows provide a separate line of context [16]. This paper instead works with structured telemetry for cellular reliability analysis, with KPI forecasting, weak degradation classification and weak alarm-candidate prioritization evaluated on a private operations slice. Section 5.7 states the limits on causality, expert RCA validation and end-to-end neural spatio-temporal graph evaluation.

The paper makes four contributions:

1. Alarm-aware KPI degradation prediction is formulated as a multi-task evaluation problem over real cellular KPI/business and alarm/fault panels.
2. A temporally admissible, causal-inspired event-token representation

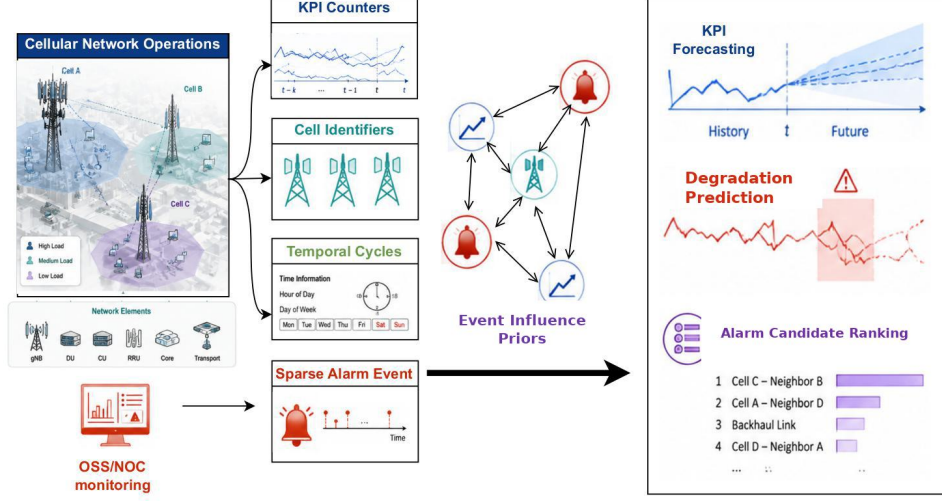


Figure 1: Overview of CET-STGL from reliability-oriented cellular network operations to analysis.

is formalised that aligns KPI patch features, alarm event features, cell/time context, train-only event influence priors and inferred cross-cell context.

3. A leakage-controlled reliability evaluation protocol is provided for 1/3/6/12-hour KPI forecasting, weak degradation classification and weak alarm-candidate prioritization.
4. The empirical findings are narrow by design: KPI-history baselines remain strong, event-intensity variants are weak standalone predictors, the evaluated lightweight CET-STGL instantiation is competitive on several primary-KPI forecasting metrics, and gains from alarm and graph features are limited and non-uniform under the current weak-label protocol.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formulates the tasks. Section 4 presents CET-STGL. Section 5 reports the experimental evaluation and limitations. Section 6 concludes the paper.

2. Related Work

2.1. Cellular KPI Forecasting and Time-Series Prediction

Mobile and wireless networking research has studied prediction as a foundation for proactive management. Recent sixth-generation (6G) surveys emphasize programmable, data-driven radio access networks for telemetry-driven control and assurance [17]. O-RAN studies connect radio access programmability with operational automation [2]. Recent RAN alarm prediction work reflects the value of modeling sparse alarms with network context [18]. Universal mobile-traffic forecasting similarly motivates learning cellular traffic dynamics [19].

General time-series forecasting now spans recurrent, convolutional, transformer, decomposition and language-model-inspired designs. TimesNet represents a recent temporal-variation modeling direction [20], while strong linear baselines question whether transformer complexity is always necessary [21]. Follow-up models explore inverted transformer structures, multi-scale mixing and probabilistic sequence pretraining [22–24]. Time-LLM and GPT4MTS adapt language-model ideas to time-series prediction [25, 26]. Donut and OmniAnomaly further motivate learning from seasonal or multivariate KPI time series in production systems [27, 28]. Cellular KPI forecasting studies mainly focus on traffic prediction or individual KPI modeling, whereas operational degradation prediction requires continuous KPI evolution, sparse fault events and cross-cell context. Existing approaches rarely address alarm-KPI coupling under enterprise monitoring conditions with incomplete topology information and expert RCA annotations. CET-STGL addresses this setting by integrating KPI forecasting, weak degradation estimation and alarm-candidate prioritization within an event-token representation framework.

2.2. Alarm Correlation, RCA and Event Influence Modeling

Alarm correlation is a classic fault-management problem in communication networks. Recent work has explored alarm-correlation rule mining for network fault management [29], whereas log-based operational diagnosis methods such as DeepLog illustrate the broader role of event sequences in detecting and diagnosing operational anomalies [30].

Recent artificial intelligence for information technology operations (AIOps) and microservice RCA studies provide methodological background for multimodal observability and graph-based diagnosis. Nezha models

multi-modal observability for interpretable root-cause analysis [31]. Causal-RCA and neural Granger causal discovery explore causal-inference-inspired localization [32, 33], and Chain-of-Event learns weighted event causal graphs for microservice incidents [34]. RCAEval and AIOpsLab highlight the need for reproducible RCA evaluation infrastructure [35, 36]. Point-process analysis provides a foundation for self-exciting event sequences [37], and Hawkes-based Granger modeling connects event intensity with temporal influence estimation [38]. Existing alarm correlation and RCA methods often focus on post-incident localization. Proactive degradation prediction instead asks whether sparse alarms add useful context before KPI deterioration. In cellular operations, alarms are often noisy, delayed or incomplete, making direct event-based prediction challenging. CET-STGL treats alarms as contextual event signals and studies their contribution through event tokens, temporal intensity and influence modeling. Ranking results are interpreted as weak-label consistency rather than validated RCA, and event influence features are causal-inspired association priors rather than proof of causality.

2.3. Spatio-Temporal Modeling and Telecom LLM Context

Spatio-temporal graph neural networks model temporal dynamics and entity dependencies. Recent traffic-forecasting work has developed delay-aware transformers for long-range propagation effects [39]. Adaptive spatio-temporal embeddings and large-scale traffic datasets support systematic evaluation of interacting entities [40, 41]. Recent KDD work on asynchronous spatio-temporal graph convolution also highlights irregular temporal structure in graph-based forecasting [42]. Existing spatio-temporal graph learning methods show the importance of interactions among network entities, but cellular degradation prediction remains limited by incomplete topology information and operational labels. CET-STGL adapts this idea to cellular monitoring by using inferred cross-cell context and train-only event influence priors. We evaluate a lightweight sklearn-based instantiation; end-to-end neural graph encoders are left for future evaluation under the same protocol.

Recent telecom and networking LLM work examines language models for standards knowledge, networking tasks and domain adaptation. TeleQnA evaluates whether LLMs capture domain-specific standards and networking concepts [43]. LLM-oriented telecom research helps position telecom-domain AI, but CET-STGL addresses a different data modality: structured operational telemetry for KPI forecasting, weak degradation classification and weak alarm-candidate prioritization on a private operations slice.

3. Problem Formulation

Let $\mathcal{C}$ denote the cell set, $\mathcal{A}$ the alarm-type set and t hourly timestamps. For each cell-time pair (c, t) , the panel contains a KPI/business vector $\mathbf{x}_{c,t} \in \mathbb{R}^M$ and an alarm/fault vector $\mathbf{a}_{c,t} \in \mathbb{R}^A$. In the evaluated anonymized MBB panel, $|\mathcal{C}| = 33$, $M = 48$ and $A = 137$, with 11 active alarm/fault columns. Given a lookback window

$$\mathcal{W}_{c,t} = \{(\mathbf{x}_{c,\tau}, \mathbf{a}_{c,\tau}) : t - L + 1 \leq \tau \leq t\}, \quad (1)$$

the objective is to predict outcomes at $h \in \{1, 3, 6, 12\}$ hours. Windows crossing the detected long time gap are removed before modeling.

For a target $(c, t + h)$, CET-STGL uses a representation

$$\mathbf{z}_{c,t,h} = \Phi_h(\mathcal{W}_{c,t}, c, t; \Theta_{\text{train}}), \quad (2)$$

where Θ_{train} contains thresholds, scalars, event priors and graph statistics fitted on the chronological training partition. The representation is temporally admissible when $\mathbf{z}_{c,t,h}$ depends only on $\mathcal{W}_{c,t}$, identifiers and time features available at t , and Θ_{train} .

3.1. KPI Forecasting

The forecasting task predicts future KPI values:

$$\hat{\mathbf{x}}_{c,t+h} = f_h(\mathbf{z}_{c,t,h}), \quad (3)$$

where f_h is the horizon-specific forecasting map. Forecasting labels are real future observations from the private cellular panel. We report aggregate metrics over all KPI/business columns and primary-KPI metrics for the paper-level comparison.

3.2. Weak Degradation Classification

The degradation task predicts a weak binary label:

$$\hat{p}_{c,t+h} = g_h(\mathbf{z}_{c,t,h}), \quad y_{c,t+h} = \mathbf{1} \left[\bigvee_{j=1}^5 q_j(c, t + h) \right]. \quad (4)$$

The five independent OR conditions q_j are high call-drop rate above $\max(1.0\%, Q_{0.99}^c)$, low Radio Resource Control (RRC) setup success rate below $\min(98.0\%, Q_{0.05}^c)$, low Evolved Radio Access Bearer (E-RAB) setup

success rate below $\min(98.0\%, Q_{0.05}^c)$, low downlink user throughput below $Q_{0.10}^c$, and low dimensionless Channel Quality Indicator (CQI) below $Q_{0.10}^c$, where Q_α^c denotes the cell-specific training quantile of the corresponding KPI. The matching full-training threshold is used when the cell-level threshold cannot be computed from at least 24 non-missing training samples or is not finite. The label is therefore a weak KPI degradation proxy rather than an expert incident label, following weak-supervision practice for indirect or noisy supervision [44, 45].

3.3. Weak Alarm-Candidate Prioritization

For degraded samples with recent alarms, the candidate set is

$$\mathcal{A}_{c,t} = \left\{ a \in \mathcal{A} : \sum_{\tau=t-L+1}^t \mathbf{1}[\mathbf{a}_{c,\tau}^{(a)} > 0] > 0 \right\}. \quad (5)$$

The weak target candidate is induced by train-only alarm-to-degradation lift and recent event intensity,

$$a_{c,t,h}^* = \arg \max_{a \in \mathcal{A}_{c,t}} \max(0, \ell_h(a)) \lambda_a(c, t), \quad (6)$$

where $\ell_h(a)$ is the train-only log-lift and $\lambda_a(c, t)$ is the decayed event intensity. A model produces a ranked list $\hat{\pi}_{c,t,h}$ over $\mathcal{A}_{c,t}$. The task evaluates weak-label consistency only and is not expert-confirmed fault localization.

3.4. Causal-Inspired Graph Boundary

The event influence graph is constructed from temporal precedence and train-only association. It is useful for organizing candidate event influence, but the observational data contain no interventions or expert causal annotations. We therefore use only causal-inspired terminology and avoid causal-effect claims.

4. CET-STGL Representation and Evaluated Instantiation

4.1. Overview

CET-STGL is a causal-inspired event-token spatio-temporal representation framework for structured cellular operations data. Heterogeneous signals

are first aligned to the same cell-time-horizon index and then passed to forecasting, classification or prioritization heads. The evaluated representation is

$$\mathbf{z}_{c,t,h} = [\phi_{c,t}^{\text{kpi}}, \phi_{c,t}^{\text{event}}, \phi_{c,t}^{\text{pos}}, \phi_{c,t,h}^{\text{prior}}, \phi_{c,t}^{\text{graph}}], \quad (7)$$

where the five blocks correspond to KPI patches, alarm events, cell/time context, event influence priors and inferred cross-cell context. We evaluate a lightweight sklearn-based instantiation to examine the representation under a leakage-controlled protocol.

The design follows three reliability-oriented constraints. KPI histories are summarized as local level and slope patches, sparse alarms are encoded by counts, decayed intensity and train-only association priors, and cross-cell context is inferred from training-period operational correlations when topology is unavailable. Compared with simple feature concatenation, CET-STGL imposes an explicit prediction-time constraint on each (c, t, h) target and uses one admissible cell-time representation across all three evaluation tasks.

For any prediction target at $(c, t + h)$, the evaluated CET-STGL feature vector is a function only of the lookback window $\mathcal{W}_{c,t}$, identifiers and time features available at t , and statistics fitted on the chronological training partition. No validation/test labels, post- t observations or future event information are used in feature construction. KPI patch features, alarm counts and decayed intensities are computed within $\mathcal{W}_{c,t}$. Degradation thresholds, feature scalers, event lifts and cross-cell graph statistics are estimated from the training partition only, and windows crossing the long time discontinuity are excluded before model fitting. This admissibility statement concerns leakage-controlled representation construction, not causal identification or performance optimality.

The model structure and the evaluated lightweight implementation are summarized in Fig. 2.

4.2. KPI, Event and Positional Features

For each KPI m in a lookback window, the sequence $x_{c,t-L+1:t}^{(m)}$ is divided into short patches and summarized by local statistics:

$$\phi_{c,t,p,m}^{\text{kpi}} = [\text{mean}, \text{slope}]_{c,t,p,m}. \quad (8)$$

The patch summaries preserve local level and slope while reducing raw sequence length. Patch-based time-series modeling provides a direct precedent for representing long temporal windows through compact local segments [46].

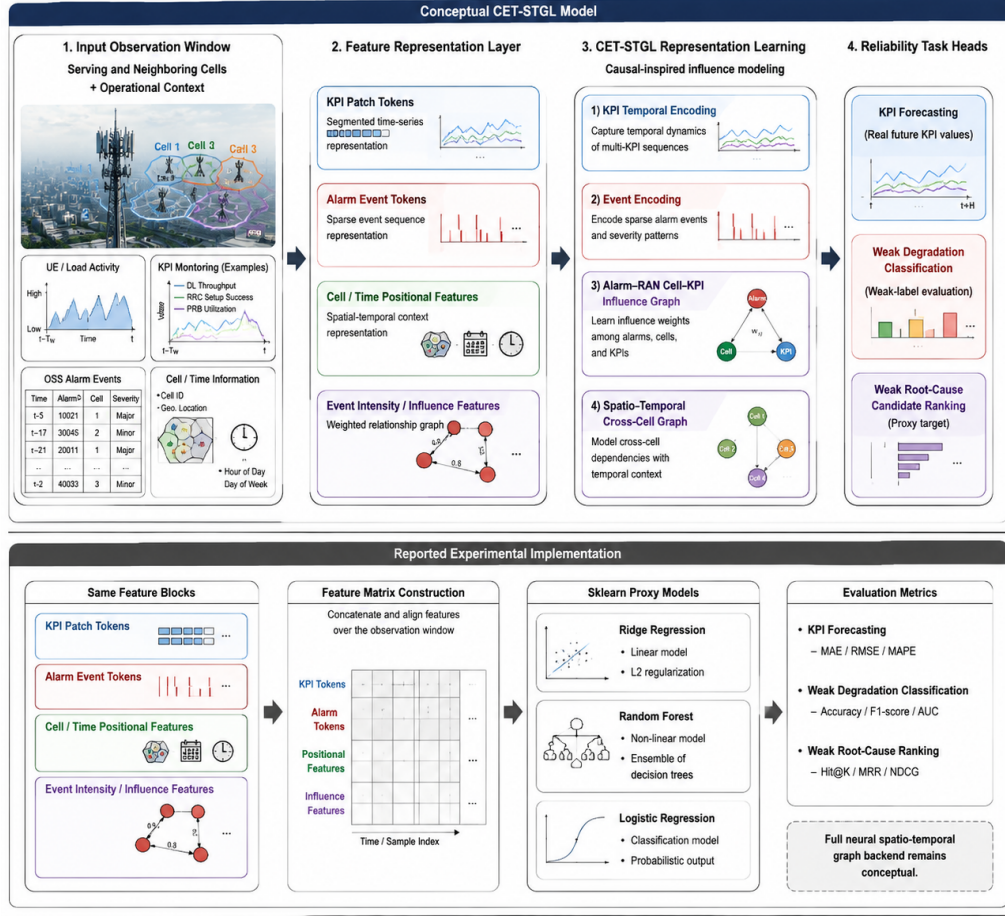


Figure 2: Conceptual structure of CET-STGL, transforming heterogeneous cellular telemetry into event-aware spatio-temporal representations with influence modeling and a lightweight sklearn-based implementation for evaluation.

Alarm/fault observations are represented as sparse event features. For alarm type a , we compute rolling counts and a decayed event intensity

$$\lambda_a(c, t) = \sum_{\tau \leq t} \mathbf{1}[a_{c,\tau} > 0] \exp\left(-\frac{t - \tau}{\eta}\right), \quad (9)$$

where η is a fixed decay scale. The term Hawkes-inspired refers to the recency-weighted form, in which recent events contribute more strongly than older events [47, 48]. In the evaluated implementation, the decayed intensity is a fixed event feature, which is appropriate for the observed sparsity: only

11 of 137 alarm/fault columns are active and only 1.62% of rows contain an active alarm.

Cell identity and time context are encoded through cell-code features and cyclical time features such as hour-of-day and day-of-week. These positional features capture persistent cell offsets and periodic operational behavior. All transformations are fitted within the chronological training protocol, and windows crossing the detected long time gap are excluded.

4.3. Event Influence Prior and Cross-Cell Context

For each alarm type and horizon, the evaluated instantiation estimates a train-only lagged lift:

$$\text{lift}_h(a) = \log \frac{P_{\text{train}}(y_{c,t+h} = 1 \mid a \in \mathcal{W}_{c,t}) + \epsilon}{P_{\text{train}}(y_{c,t+h} = 1) + \epsilon}, \quad (10)$$

where P_{train} denotes a training-set probability and ϵ is a smoothing constant. The lift is combined with recent intensity to form $\phi_{c,t,h}^{\text{prior}}$. This prior encodes training-period association between alarms and weak degradation; it is causal-inspired but does not estimate a causal effect.

The evaluated panel does not include verified physical topology. We infer a cross-cell context graph $G_{\text{cell}} = (\mathcal{C}, E)$ from training-period KPI correlations. For cell c , neighbor context is summarized as

$$\phi_{c,t}^{\text{graph}} = \frac{1}{|\mathcal{N}(c)|} \sum_{c' \in \mathcal{N}(c)} \mathbf{u}_{c',t}, \quad (11)$$

where $\mathcal{N}(c)$ is the inferred neighbor set and $\mathbf{u}_{c',t}$ denotes selected KPI context for cell c' . The inferred graph is a statistical context graph, not a physical deployment graph. Training-derived cross-cell context follows the broader idea of using graph structure or adaptive spatial dependencies for time-series forecasting, while remaining explicitly statistical in the present dataset [49, 50].

4.4. Evaluated Prediction Heads

The evaluated heads are

$$\hat{\mathbf{x}}_{c,t+h} = f_h^{\text{reg}}(\mathbf{z}_{c,t,h}), \quad \hat{p}_{c,t+h} = g_h^{\text{cls}}(\mathbf{z}_{c,t,h}), \quad \hat{\pi}_{c,t,h} = \text{rank}_{a \in \mathcal{A}_{c,t}} s_h(a, c, t). \quad (12)$$

The implementation uses Ridge regression for multi-output KPI forecasting, a linear logistic classifier trained using stochastic gradient descent for weak degradation classification, and event-lift/intensity scoring for weak alarm-candidate prioritization. End-to-end neural spatio-temporal graph evaluation remains future work under the same leakage-controlled protocol.

5. Experimental Evaluation

5.1. Data and Evaluation Protocol

The evaluation uses a real-world MBB operational dataset provided by Huawei Technologies. The source collection covers approximately 500 anonymized cellular sites, while the aligned panel evaluated in this study contains 33 anonymized cell entities after task-specific panel construction. The processed panel comprises 103,983 cell-time instances, 3,151 unique hourly timestamps, 48 KPI/business counters and 137 alarm/fault columns. Only 11 alarm/fault columns are active, and 1.62% of rows contain an active alarm. The panel contains one 21 days 01:00:00 time discontinuity; forecasting windows crossing this gap are removed. Dataset dimensions and feature sparsity characteristics are detailed in Table 1.

All experiments use chronological splitting with 70% training, 15% validation and 15% testing. Feature scalers, degradation thresholds, inferred cell interaction graphs and event influence priors are fitted on the training partition only. Models are evaluated at four horizons ($h \in \{1, 3, 6, 12\}$ hours). The primary task is continuous future KPI value forecasting. Secondary tasks are weak KPI degradation classification based on train-derived threshold violations and weak alarm-candidate prioritization based on event intensity and lift.

5.2. Compared Models and Metrics

The runnable comparison includes Ridge regression for forecasting [51], a linear logistic classifier for weak degradation classification, corresponding no-alarm variants, recency-weighted event-intensity baselines, event-feature scoring baselines, CET-STGL-sklearn, its ablations, and two fixed neural sequence baselines: Long Short-Term Memory (LSTM) [52] and Temporal Convolutional Network (TCN) [53]. The sklearn-compatible implementation follows standard Python machine-learning tooling [54]. "Ridge/Logistic" is used as a compact cross-task label in the result tables: forecasting uses Ridge

Table 1: Dataset statistics for the anonymized Huawei MBB operations data and evaluated cell-time panel.

Item	Value
Collection scope	Approximately 500 MBB cellular sites in the source collection
Rows	103,983
Original columns	187
Anonymized cell entities in evaluated panel	33
Unique hourly timestamps	3,151
Observed time structure	3,151 unique hourly timestamps in the evaluated panel; exact site-identifiable information is not used
KPI/business columns	48
Alarm/fault columns	137
Active alarm/fault columns	11
Rows with active alarms	1,681 (1.62%)
Missing values	None detected
Detected time discontinuity	One 21 days 01:00:00 gap; crossing windows removed

regression, whereas classification uses the SGD-trained linear logistic classifier. LSTM and TCN use the same 24-hour observation window, chronological split, train-only scaling, active alarm inputs and fixed hyperparameters without search. They are sequence-model comparators rather than full neural spatio-temporal graph implementations.

Continuous KPI forecasting is measured using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) with a numerical floor $\max(|y|, 1.0)$ to stabilize zero-value division. Degradation classification performance is quantified via Area Under the Precision-Recall Curve (AUPRC), Area Under the ROC Curve (AUROC), F1-score, Precision, and Recall. Weak alarm-candidate ranking consistency is evaluated using the hit rate at cutoff K , denoted $\text{Hit}(K)$ for $K \in \{1, 3, 5\}$, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain at cutoff 5, denoted $\text{nDCG}(5)$. Here, $\text{Hit}(K)$ means that

the weak target alarm appears among the top- K ranked candidates, and $\text{nDCG}(5)$ is computed after truncating the ranked list at five candidates.

5.3. KPI Forecasting Results

Continuous KPI forecasting performance across horizons is reported in Table 2, with the main CET-STGL, classical and event-based trends visualized in Fig. 3. CET-STGL-sklearn is competitive with the feature-engineered Ridge baseline on the primary-KPI subset. At $h = 1$ h, CET-STGL-sklearn obtains MAE/RMSE values of 2.183/4.635, compared with 2.219/4.698 for Ridge/Logistic. At $h = 3$ h and $h = 6$ h, CET-STGL-sklearn remains lower in MAE than Ridge/Logistic. At $h = 12$ h, the two models are close: CET-STGL-sklearn has slightly lower MAE, while Ridge/Logistic has slightly lower RMSE.

The no-alarm CET-STGL ablation achieves lower MAE and RMSE than the full CET-STGL instantiation at all four horizons. The pattern is consistent with the high sparsity of alarm observations: active alarms appear in only 1.62% of rows, while routine KPI variation is largely described by continuous temporal and cell-specific patterns. From a reliability perspective, the ablation indicates that sparse alarms should be treated as contextual evidence rather than universal predictors of routine KPI evolution. Standalone event-intensity baselines remain weak for continuous forecasting, with Hawkes-only producing an $h = 1$ h MAE/RMSE of 4.578/7.981. The fixed LSTM baseline has higher primary-KPI MAE/RMSE than CET-STGL-sklearn and Ridge/Logistic at all horizons, and the fixed TCN baseline is weaker under the same protocol.

5.4. Weak Degradation Classification Results

Weak degradation classification performance is summarized in Table 3, while Fig. 4 gives the corresponding CET-STGL, classical and event-based comparisons. At $h = 1$ h, CET-STGL-sklearn obtains an AUPRC of 0.466 and F1-score of 0.479, performing comparably to the feature-engineered linear logistic baselines (AUPRC 0.471–0.476). Across longer horizons, AUPRC values among the feature-engineered baselines remain relatively close, whereas threshold-dependent F1, Precision and Recall vary substantially across models and horizons. LSTM provides a useful neural sequence comparator and is competitive on AUPRC at some horizons, but it does not dominate the linear and CET-STGL variants across metrics.

Table 2: Primary-KPI forecasting summary for runnable models.

Model	H	MAE	RMSE	MAPE
CET-STGL without alarm tokens	1	2.157	4.602	42.411
CET-STGL-sklearn	1	2.183	4.635	42.642
Event-attention-only	1	4.630	8.276	82.519
Hawkes-only	1	4.578	7.981	82.261
LSTM	1	3.774	6.606	68.954
TCN	1	13.360	51.673	131.255
Ridge/Logistic	1	2.219	4.698	43.599
Ridge/Logistic without alarms	1	2.203	4.671	43.382
CET-STGL without alarm tokens	3	2.490	4.974	46.807
CET-STGL-sklearn	3	2.537	5.083	47.162
Event-attention-only	3	4.635	8.326	82.776
Hawkes-only	3	4.581	8.038	82.515
LSTM	3	3.822	6.907	72.139
TCN	3	6.580	16.235	116.662
Ridge/Logistic	3	3.071	5.684	56.382
Ridge/Logistic without alarms	3	3.036	5.586	56.015
CET-STGL without alarm tokens	6	3.145	5.605	56.843
CET-STGL-sklearn	6	3.197	5.853	57.264
Event-attention-only	6	4.629	8.277	83.117
Hawkes-only	6	4.578	7.873	82.871
LSTM	6	3.991	7.053	73.741
TCN	6	7.660	19.911	121.740
Ridge/Logistic	6	3.338	5.940	61.590
Ridge/Logistic without alarms	6	3.318	5.884	61.272
CET-STGL without alarm tokens	12	2.515	5.051	47.283
CET-STGL-sklearn	12	2.604	5.442	47.682
Event-attention-only	12	4.671	8.622	83.341
Hawkes-only	12	4.572	7.888	83.061
LSTM	12	3.892	7.066	73.416
TCN	12	11.522	41.975	124.861
Ridge/Logistic	12	2.640	5.391	49.140
Ridge/Logistic without alarms	12	2.593	5.135	48.806

Event-only models exhibit limited discriminative power for threshold-based degradation prediction, with Hawkes-only and Event-attention-only attaining lower AUPRC scores of 0.247 and 0.257 at $h = 1$ h, respectively. Sparse network alarms are weak standalone predictors under the present protocol.

5.5. Weak Alarm-Candidate Prioritization Results

Weak alarm-candidate ranking consistency is evaluated in Table 4 and Fig. 6. Both the target candidate and the ranking score are derived from association-based weak rules using event intensity and train-only lift. Values of 1.000 therefore indicate agreement with the weak protocol, not verified operational fault localization. The Hawkes-only and no-dynamic-graph rows

Table 3: Weak KPI degradation classification summary.

Model	H	AUPRC	F1	Prec.	Recall
CET-STGL without alarm tokens	1	0.470	0.361	0.221	0.975
CET-STGL-sklearn	1	0.466	0.479	0.395	0.608
Event-attention-only	1	0.257	0.331	0.226	0.620
Hawkes-only	1	0.247	0.327	0.218	0.656
LSTM	1	0.448	0.404	0.277	0.748
TCN	1	0.270	0.389	0.277	0.652
Ridge/Logistic	1	0.471	0.477	0.418	0.556
Ridge/Logistic without alarms	1	0.476	0.480	0.396	0.610
CET-STGL without alarm tokens	3	0.399	0.348	0.520	0.262
CET-STGL-sklearn	3	0.375	0.147	0.570	0.084
Event-attention-only	3	0.249	0.331	0.218	0.684
Hawkes-only	3	0.233	0.326	0.218	0.649
LSTM	3	0.370	0.390	0.268	0.717
TCN	3	0.220	0.334	0.226	0.640
Ridge/Logistic	3	0.341	0.320	0.191	0.996
Ridge/Logistic without alarms	3	0.336	0.212	0.445	0.139
CET-STGL without alarm tokens	6	0.334	0.331	0.200	0.975
CET-STGL-sklearn	6	0.318	0.389	0.284	0.618
Event-attention-only	6	0.215	0.352	0.245	0.623
Hawkes-only	6	0.294	0.318	0.215	0.611
LSTM	6	0.350	0.368	0.246	0.727
TCN	6	0.199	0.313	0.211	0.605
Ridge/Logistic	6	0.309	0.315	0.402	0.259
Ridge/Logistic without alarms	6	0.284	0.363	0.245	0.698
CET-STGL without alarm tokens	12	0.377	0.334	0.201	0.989
CET-STGL-sklearn	12	0.373	0.324	0.193	0.997
Event-attention-only	12	0.240	0.340	0.238	0.590
Hawkes-only	12	0.247	0.322	0.219	0.608
LSTM	12	0.380	0.376	0.262	0.666
TCN	12	0.190	0.289	0.196	0.548
Ridge/Logistic	12	0.381	0.334	0.201	0.983
Ridge/Logistic without alarms	12	0.396	0.335	0.202	0.983

act as diagnostic checks because they retain event recency but remove the full association-derived scoring structure, yielding Hit(1) values near 0.80 and MRR values near 0.89.

5.6. Ablation and Component Analysis

Ablation deltas relative to the full CET-STGL instantiation are reported in Table 5 and Fig. 5. Removing alarm tokens lowers average forecasting MAE in the reported primary-KPI summary. In the evaluated slice, alarm and graph features do not improve aggregate forecasting consistently, although alarm information remains relevant to the weak association-based prioritization protocol. The ablation evidence therefore supports the methodological separation between continuous KPI dynamics, sparse event context and association-based prioritization.

Table 4: Weak alarm-candidate ranking consistency summary.

Model	H	N	Hit(1)	MRR	nDCG(5)
CET-STGL without dynamic event graph	1	2682	0.808	0.895	0.920
CET-STGL-sklearn	1	2682	1.000	1.000	1.000
Event-attention-only	1	611	1.000	1.000	1.000
Hawkes-only	1	2682	0.808	0.895	0.920
Ridge/Logistic	1	611	0.930	0.960	0.971
CET-STGL without dynamic event graph	3	2671	0.807	0.894	0.919
CET-STGL-sklearn	3	2671	1.000	1.000	1.000
Event-attention-only	3	609	1.000	1.000	1.000
Hawkes-only	3	2671	0.807	0.894	0.919
Ridge/Logistic	3	609	0.923	0.958	0.969
CET-STGL without dynamic event graph	6	2645	0.805	0.893	0.918
CET-STGL-sklearn	6	2645	1.000	1.000	1.000
Event-attention-only	6	614	1.000	1.000	1.000
Hawkes-only	6	2645	0.805	0.893	0.918
Ridge/Logistic	6	614	0.917	0.955	0.967
CET-STGL without dynamic event graph	12	2617	0.804	0.892	0.918
CET-STGL-sklearn	12	2617	1.000	1.000	1.000
Event-attention-only	12	649	1.000	1.000	1.000
Hawkes-only	12	2617	0.804	0.892	0.918
Ridge/Logistic	12	649	0.928	0.962	0.972

Table 5: Ablation deltas averaged over horizons relative to CET-STGL-sklearn.

Ablation row	Removed component	Δ MAE	Δ AUPRC	Δ MRR
CET-STGL without alarm tokens	no_alarm	-0.054	0.012	
CET-STGL without cross-cell graph	no_cross_cell_graph	-0.003	-0.000	0.000
CET-STGL without dynamic event graph	no_dynamic_graph	0.000	0.001	-0.107
CET-STGL without Hawkes intensity	no_hawkes_intensity	0.000	0.001	0.000
CET-STGL without tokenization	no_tokenization	0.200	-0.017	0.000

Component-wise ablations further clarify the role of specific structural choices. Removing tokenization increases average forecasting MAE, supporting the use of local level and slope summaries for KPI histories. Removing the dynamic event graph reduces consistency with the association-derived ranking rule, as expected. The result reflects protocol consistency rather than independent validation of operational fault-localization utility.

5.7. Limitations, Data Availability and Reproducibility

The study is limited to the evaluated panel and weak labels. KPI histories and cell-time context carry most of the forecasting signal; alarm and inferred graph features yield limited, non-uniform aggregate gains. Weak alarm-candidate ranking is a proxy consistency check rather than expert-confirmed RCA, and the event influence graph captures observational association rather than estimated causal effects.

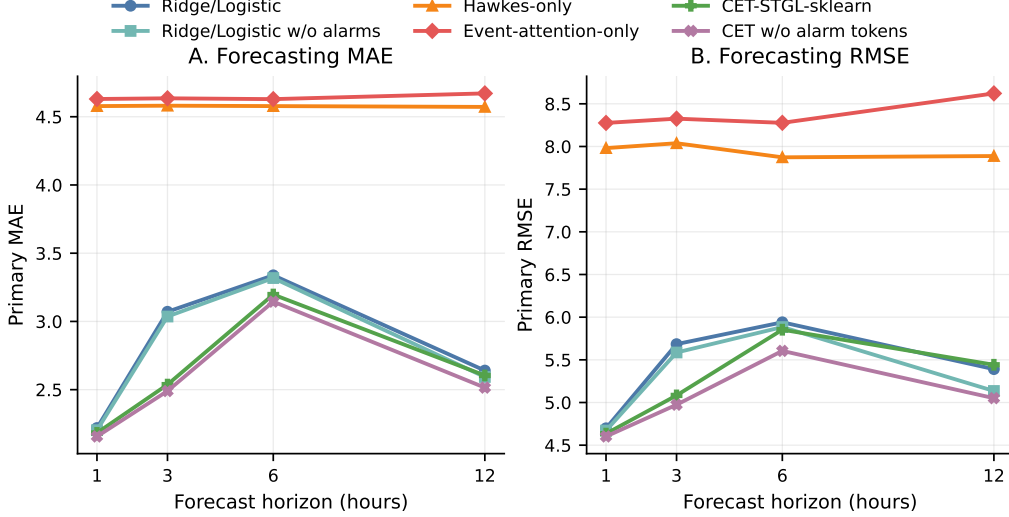


Figure 3: Primary-KPI forecasting error versus prediction horizon.

The chronological split tests future-time generalization within observed cells, not transfer to unseen regions, networks or deployments. End-to-end neural spatio-temporal graph evaluation is left to future work. Raw Huawei MBB records cannot be released because they include confidential operational KPI and alarm information. Code, sanitized schemas, synthetic validation data and experiment scripts are available at <https://github.com/DunLi-Tsinghua/cet-stgl-cellular-kpi>.

6. Conclusion

This paper presents CET-STGL, a causal-inspired event-token spatio-temporal framework for alarm-aware KPI degradation prediction in cellular networks. The main methodological contribution is a temporally admissible representation pipeline that organizes KPI histories, sparse alarm events, cell/time context, train-only event influence priors and inferred cross-cell context under a common reliability evaluation protocol. On anonymized Huawei MBB operations data, the evaluated lightweight instantiation is competitive with classical baselines on several primary-KPI forecasting metrics, while KPI-history and no-alarm variants remain strong. Alarm and graph features do not improve aggregate forecasting consistently in the evaluated slice, although alarm information remains relevant to the weak association-based

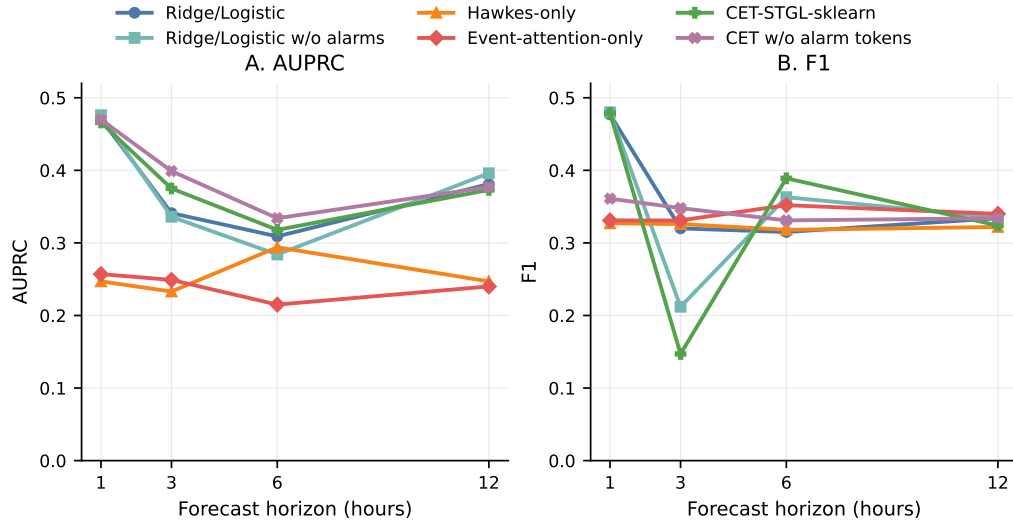


Figure 4: KPI degradation classification performance versus prediction horizon.

prioritization protocol. The ranking task measures weak-label consistency rather than expert-confirmed fault localization. Future work should evaluate additional operational slices, incorporate verified topology where available, add expert incident annotations and test end-to-end neural spatio-temporal graph instantiations under the same protocol.

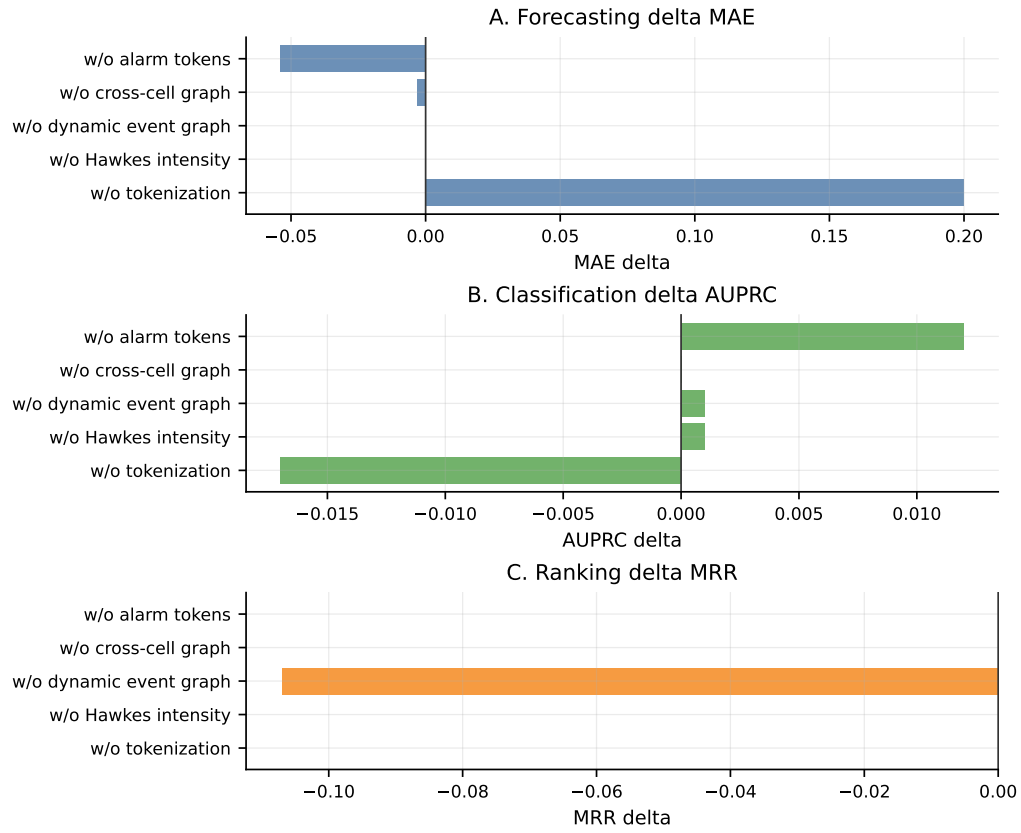


Figure 5: Ablation deltas relative to CET-STGL-sklearn.

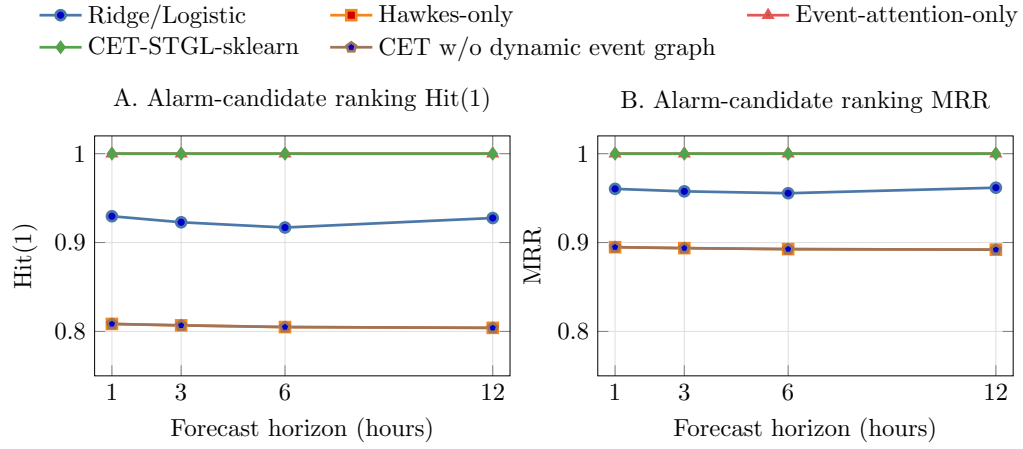


Figure 6: Weak alarm-candidate ranking consistency versus prediction horizon.

Appendix A. Full Experimental Tables

The main paper reports compact summary tables for readability. The appendix retains the full runnable model-by-horizon tables generated from the existing paper artifacts.

Appendix A.1. Full Forecasting Results

Table A.1: KPI forecasting results over runnable models and horizons.

Model	H	P-MAE	P-RMSE	P-MAPE	All-MAE	All-RMSE
CET-STGL without alarm tokens	1	2.1569	4.6016	42.4111	30.1717	53.1923
CET-STGL without cross-cell graph	1	2.1791	4.6351	42.7362	31.1928	62.4168
CET-STGL without dynamic event graph	1	2.1831	4.6351	42.6425	31.2506	62.1414
CET-STGL without Hawkes intensity	1	2.1831	4.6351	42.6425	31.2506	62.1414
CET-STGL without tokenization	1	2.2244	4.7059	43.4812	31.6755	61.3319
CET-STGL-sklearn	1	2.1830	4.6351	42.6425	31.2510	62.1427
Event-attention-only	1	4.6305	8.2757	82.5194	163.9535	271.1688
Hawkes-only	1	4.5779	7.9814	82.2610	160.0211	227.2084
LSTM	1	3.7740	6.6063	68.9540	77.4459	123.5975
TCN	1	13.3600	51.6732	131.2548	343.2180	1287.6357
Ridge/Logistic	1	2.2191	4.6984	43.5993	31.4426	58.4626
Ridge/Logistic without alarms	1	2.2034	4.6712	43.3820	30.6931	53.7885
CET-STGL without alarm tokens	3	2.4904	4.9744	46.8069	39.4850	67.8885
CET-STGL without cross-cell graph	3	2.5354	5.0786	47.2707	42.1875	103.4692
CET-STGL without dynamic event graph	3	2.5368	5.0829	47.1621	42.1587	102.7291
CET-STGL without Hawkes intensity	3	2.5368	5.0829	47.1621	42.1587	102.7291
CET-STGL without tokenization	3	3.0549	5.6840	56.1056	50.7988	101.8653
CET-STGL-sklearn	3	2.5367	5.0829	47.1624	42.1585	102.7298
Event-attention-only	3	4.6346	8.3256	82.7764	164.6453	289.4013
Hawkes-only	3	4.5809	8.0385	82.5149	159.7122	227.2520
LSTM	3	3.8216	6.9066	72.1385	80.1701	132.0877
TCN	3	6.5805	16.2346	116.6616	339.5963	1156.6342
Ridge/Logistic	3	3.0709	5.6842	56.3815	50.7191	93.6261
Ridge/Logistic without alarms	3	3.0361	5.5858	56.0146	48.9354	78.7454
CET-STGL without alarm tokens	6	3.1447	5.6052	56.8431	50.7871	82.1649
CET-STGL without cross-cell graph	6	3.1904	5.8644	57.2840	54.4509	133.8707
CET-STGL without dynamic event graph	6	3.1970	5.8527	57.2636	54.4014	132.2650

Model		H	P-MAE	P-RMSE	P-MAPE	All-MAE	All-RMSE
CET-STGL	without	6	3.1970	5.8527	57.2636	54.4014	132.2650
Hawkes intensity							
CET-STGL	without tok-	6	3.3631	6.0530	61.5916	65.4517	131.9830
enization							
CET-STGL-sklearn		6	3.1971	5.8527	57.2638	54.4004	132.2636
Event-attention-only		6	4.6287	8.2769	83.1170	164.8499	295.9310
Hawkes-only		6	4.5785	7.8735	82.8707	159.5017	225.5720
LSTM		6	3.9908	7.0530	73.7409	80.0465	130.2718
TCN		6	7.6596	19.9105	121.7399	266.1071	777.7711
Ridge/Logistic		6	3.3380	5.9402	61.5904	65.1005	118.3171
Ridge/Logistic	without	6	3.3177	5.8838	61.2722	62.6208	97.5729
alarms							
CET-STGL	without alarm	12	2.5147	5.0508	47.2830	42.0699	74.8636
tokens							
CET-STGL	without cross-	12	2.6040	5.4528	47.7833	46.2547	127.1682
cell graph							
CET-STGL	without dy-	12	2.6041	5.4421	47.6820	46.4562	126.0803
amic event graph							
CET-STGL	without	12	2.6041	5.4421	47.6820	46.4562	126.0803
Hawkes intensity							
CET-STGL	without tok-	12	2.6796	5.5472	49.0619	47.8638	124.7888
enization							
CET-STGL-sklearn		12	2.6041	5.4421	47.6819	46.4562	126.0802
Event-attention-only		12	4.6714	8.6216	83.3406	164.9266	287.0291
Hawkes-only		12	4.5719	7.8884	83.0608	160.6519	238.1456
LSTM		12	3.8917	7.0664	73.4162	82.1490	132.8829
TCN		12	11.5222	41.9747	124.8608	332.8840	1110.3004
Ridge/Logistic		12	2.6400	5.3913	49.1403	47.3654	122.6165
Ridge/Logistic	without	12	2.5928	5.1347	48.8058	43.2913	76.8817
alarms							

Appendix A.2. Full Weak Degradation Classification Results

Table A.2: Weak KPI degradation classification results.

Model	H	AUPRC	F1	Prec.	Recall	AUROC
CET-STGL without alarm	1	0.4702	0.3609	0.2215	0.9746	0.7975
tokens						
CET-STGL without cross-	1	0.4482	0.3980	0.5244	0.3207	0.7827
cell graph						
CET-STGL without dy-	1	0.4662	0.4784	0.3948	0.6070	0.7943
amic event graph						
CET-STGL without	1	0.4662	0.4785	0.3947	0.6074	0.7944
Hawkes intensity						
CET-STGL without tok-	1	0.4578	0.4583	0.4634	0.4534	0.7892
enization						
CET-STGL-sklearn	1	0.4663	0.4787	0.3949	0.6078	0.7944
Event-attention-only	1	0.2573	0.3310	0.2257	0.6201	0.6358
Hawkes-only	1	0.2465	0.3274	0.2181	0.6562	0.6314
LSTM	1	0.4477	0.4042	0.2769	0.7479	0.7553
TCN	1	0.2699	0.3891	0.2773	0.6518	0.6846
Ridge/Logistic	1	0.4709	0.4774	0.4182	0.5563	0.7946

Model		H	AUPRC	F1	Prec.	Recall	AUROC
Ridge/Logistic	without	1	0.4756	0.4801	0.3956	0.6104	0.7971
alarms							
CET-STGL	without alarm	3	0.3990	0.3481	0.5197	0.2617	0.7587
tokens							
CET-STGL	without cross-	3	0.3870	0.4010	0.2700	0.7787	0.7527
cell graph							
CET-STGL	without dy-	3	0.3760	0.1475	0.5751	0.0846	0.7430
amic event graph							
CET-STGL	without	3	0.3759	0.1476	0.5780	0.0846	0.7429
Hawkes intensity							
CET-STGL	without tok-	3	0.3419	0.3121	0.1849	0.9985	0.7236
enization							
CET-STGL-sklearn		3	0.3752	0.1468	0.5696	0.0842	0.7427
Event-attention-only		3	0.2494	0.3310	0.2183	0.6836	0.6328
Hawkes-only		3	0.2331	0.3265	0.2181	0.6492	0.6230
LSTM		3	0.3703	0.3899	0.2678	0.7166	0.7207
TCN		3	0.2198	0.3340	0.2260	0.6402	0.6193
Ridge/Logistic		3	0.3409	0.3201	0.1907	0.9959	0.7231
Ridge/Logistic	without	3	0.3365	0.2118	0.4454	0.1389	0.7256
alarms							
CET-STGL	without alarm	6	0.3344	0.3313	0.1996	0.9750	0.7167
tokens							
CET-STGL	without cross-	6	0.3195	0.3318	0.1997	0.9792	0.7178
cell graph							
CET-STGL	without dy-	6	0.3203	0.3894	0.2853	0.6129	0.7131
amic event graph							
CET-STGL	without	6	0.3203	0.3892	0.2849	0.6136	0.7131
Hawkes intensity							
CET-STGL	without tok-	6	0.3026	0.3033	0.3729	0.2556	0.6842
enization							
CET-STGL-sklearn		6	0.3181	0.3887	0.2835	0.6181	0.7127
Event-attention-only		6	0.2146	0.3523	0.2455	0.6234	0.6165
Hawkes-only		6	0.2940	0.3183	0.2152	0.6113	0.6326
LSTM		6	0.3505	0.3678	0.2462	0.7270	0.7024
TCN		6	0.1991	0.3129	0.2111	0.6045	0.5831
Ridge/Logistic		6	0.3087	0.3152	0.4025	0.2590	0.6909
Ridge/Logistic	without	6	0.2840	0.3625	0.2448	0.6983	0.6830
alarms							
CET-STGL	without alarm	12	0.3769	0.3337	0.2007	0.9885	0.7526
tokens							
CET-STGL	without cross-	12	0.3777	0.3213	0.1915	0.9981	0.7499
cell graph							
CET-STGL	without dy-	12	0.3728	0.3243	0.1937	0.9966	0.7463
amic event graph							
CET-STGL	without	12	0.3729	0.3243	0.1937	0.9966	0.7463
Hawkes intensity							
CET-STGL	without tok-	12	0.3641	0.3633	0.4503	0.3045	0.7382
enization							
CET-STGL-sklearn		12	0.3732	0.3240	0.1934	0.9966	0.7465
Event-attention-only		12	0.2405	0.3396	0.2383	0.5904	0.6332
Hawkes-only		12	0.2467	0.3221	0.2190	0.6083	0.6107
LSTM		12	0.3801	0.3763	0.2622	0.6664	0.7101
TCN		12	0.1896	0.2885	0.1958	0.5480	0.5695
Ridge/Logistic		12	0.3806	0.3339	0.2011	0.9828	0.7413

Model		H	AUPRC	F1	Prec.	Recall	AUROC
Ridge/Logistic alarms	without	12	0.3959	0.3348	0.2018	0.9828	0.7420

Appendix A.3. Full Weak Ranking Results

Table A.3: Weak alarm-candidate ranking consistency.

Model		H	N	Hit(1)	Hit(3)	Hit(5)	MRR	nDCG(5)
CET-STGL alarm tokens	without	1	0					
CET-STGL cross-cell graph	without	1	2682	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL without dynamic event graph	without	1	2682	0.8084	0.9952	0.9959	0.8948	0.9204
CET-STGL Hawkes intensity	without	1	611	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL without tokenization		1	2682	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn		1	2682	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only		1	611	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only		1	2682	0.8084	0.9952	0.9959	0.8948	0.9204
Ridge/Logistic		1	611	0.9296	1.0000	1.0000	0.9604	0.9706
Ridge/Logistic alarms	without	1	0					
CET-STGL alarm tokens	without	3	0					
CET-STGL cross-cell graph	without	3	2671	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL without dynamic event graph	without	3	2671	0.8068	0.9944	0.9951	0.8937	0.9193
CET-STGL Hawkes intensity	without	3	609	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL without tokenization		3	2671	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn		3	2671	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only		3	609	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only		3	2671	0.8068	0.9944	0.9951	0.8937	0.9193
Ridge/Logistic		3	609	0.9228	1.0000	1.0000	0.9576	0.9685
Ridge/Logistic alarms	without	3	0					
CET-STGL alarm tokens	without	6	0					
CET-STGL cross-cell graph	without	6	2645	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL without dynamic event graph	without	6	2645	0.8049	0.9940	0.9947	0.8926	0.9183
CET-STGL Hawkes intensity	without	6	614	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL without tokenization		6	2645	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn		6	2645	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only		6	614	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only		6	2645	0.8049	0.9940	0.9947	0.8926	0.9183

Model		H	N	Hit(1)	Hit(3)	Hit(5)	MRR	nDCG(5)
Ridge/Logistic		6	614	0.9169	1.0000	1.0000	0.9555	0.9670
Ridge/Logistic	without	6	0					
alarms								
CET-STGL	without	12	0					
alarm tokens								
CET-STGL	without	12	2617	1.0000	1.0000	1.0000	1.0000	1.0000
cross-cell graph								
CET-STGL	without dy-	12	2617	0.8040	0.9935	0.9943	0.8920	0.9177
namic event graph								
CET-STGL	without	12	649	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes intensity								
CET-STGL	without tok-	12	2617	1.0000	1.0000	1.0000	1.0000	1.0000
enization								
CET-STGL-sklearn		12	2617	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only		12	649	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only		12	2617	0.8040	0.9935	0.9943	0.8920	0.9177
Ridge/Logistic		12	649	0.9276	1.0000	1.0000	0.9617	0.9717
Ridge/Logistic	without	12	0					
alarms								

Appendix A.4. Full Ablation Delta Table

Table A.4: Ablation deltas relative to CET-STGL-sklearn.

Ablation row		Removed	H	Δ P-MAE	Δ AUPRC	Δ MRR
CET-STGL	without	no_alarm	1	-0.0262	0.0039	
alarm tokens						
CET-STGL	without	no_alarm	3	-0.0463	0.0238	
alarm tokens						
CET-STGL	without	no_alarm	6	-0.0524	0.0162	
alarm tokens						
CET-STGL	without	no_alarm	12	-0.0893	0.0037	
alarm tokens						
CET-STGL	without	no_cross	1	-0.0039	-0.0181	0.0000
cross-cell graph		cell_graph				
CET-STGL	without	no_cross	3	-0.0014	0.0118	0.0000
cross-cell graph		cell_graph				
CET-STGL	without	no_cross	6	-0.0066	0.0014	0.0000
cross-cell graph		cell_graph				
CET-STGL	without	no_cross	12	-0.0001	0.0045	0.0000
cross-cell graph		cell_graph				
CET-STGL	without dy-	no_dynamic	1	0.0001	-0.0001	-0.1052
dynamic event graph		graph				
CET-STGL	without dy-	no_dynamic	3	0.0000	0.0007	-0.1063
dynamic event graph		graph				
CET-STGL	without dy-	no_dynamic	6	-0.0001	0.0022	-0.1074
dynamic event graph		graph				
CET-STGL	without dy-	no_dynamic	12	-0.0000	-0.0004	-0.1080
dynamic event graph		graph				

Ablation row			Removed	H	Δ P-MAE	Δ AUPRC	Δ MRR
CET-STGL	without	no_hawkes	intensity	1	0.0001	-0.0001	0.0000
Hawkes intensity							
CET-STGL	without	no_hawkes	intensity	3	0.0000	0.0007	0.0000
Hawkes intensity							
CET-STGL	without	no_hawkes	intensity	6	-0.0001	0.0022	0.0000
Hawkes intensity							
CET-STGL	without	no_hawkes	intensity	12	-0.0000	-0.0003	0.0000
Hawkes intensity							
CET-STGL	without tok-	no_tokenization		1	0.0414	-0.0085	0.0000
enization							
CET-STGL	without tok-	no_tokenization		3	0.5181	-0.0333	0.0000
enization							
CET-STGL	without tok-	no_tokenization		6	0.1660	-0.0155	0.0000
enization							
CET-STGL	without tok-	no_tokenization		12	0.0755	-0.0091	0.0000
enization							

Appendix B. Nomenclature

Table B.1: Nomenclature of Acronyms and Symbols.

Item	Description
Acronyms	
AIOps	Artificial Intelligence for IT Operations
AUPRC	Area Under the Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
CET-STGL	Causal-Inspired Event-Token Spatio-Temporal Framework
KPI	Key Performance Indicator
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MBB	Mobile Broadband
MRR	Mean Reciprocal Rank
nDCG	Normalized Discounted Cumulative Gain
O-RAN	Open Radio Access Network
RCA	Root-Cause Analysis
RMSE	Root Mean Square Error
Symbols	
$\mathcal{C}$	Set of evaluated cellular sites
c, t	Indices for cell entity and hourly timestamp
h	Forecasting horizon in hours ($h \in \{1, 3, 6, 12\}$)
L	Lookback window length
$\mathbf{x}_{c,t}$	KPI and business counter vector at cell c , time t
$\mathbf{a}_{c,t}$	Alarm and fault vector at cell c , time t
$\mathcal{W}_{c,t}$	Historical lookback window for a given cell and time
M, A	Dimensionality of KPI vector and alarm vector, respectively
f_h, g_h, r_h	Mapping functions for forecasting, classification, and ranking tasks
$\hat{\mathbf{x}}_{c,t+h}$	Predicted KPI vector at horizon h
$y_{c,t+h}, \hat{y}_{c,t+h}$	Ground-truth weak binary degradation indicator and predicted probability
$\mathcal{A}_{c,t}$	Set of candidate alarm types for ranking
$s_h(a, c, t)$	Ranking score assigned to alarm candidate a
$\phi_{c,t,p,m}^{\text{kpi}}$	Extracted patch-based feature summary for KPI m
$\lambda_a(c, t)$	Decayed event intensity for alarm type a
η	Fixed decay scale for event intensity
$\text{lift}_h(a)$	Train-only event influence prior (lagged lift) for alarm a
P_{train}	Probability estimated within the training set
G_{cell}	Inferred cross-cell statistical context graph
E	Edges within the cross-cell graph G_{cell}

Appendix C. Data Availability and Reproducibility

The underlying Huawei MBB operational records are anonymized enterprise data and cannot be publicly released because they contain confidential network performance and alarm information. To support reproducibility within these restrictions, the project repository provides the source code, sanitized schema, synthetic examples and artifacts used to generate the

manuscript tables and figures. Exact metric reproduction requires authorized access to the raw operational records and the chronological train-only fitting protocol described in Section 5.1.

References

- [1] X. Lin, L. Kundu, C. Dick, S. Velayutham, Embracing AI in 5G-advanced toward 6G: A joint 3GPP and O-RAN perspective, *IEEE Communications Standards Magazine* 7 (4) (2023) 76–83. doi:10.1109/MCOMSTD.0005.2200070.
URL <https://doi.org/10.1109/MCOMSTD.0005.2200070>
- [2] M. Polese, L. Bonati, S. D’Oro, S. Basagni, T. Melodia, Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges, *IEEE Communications Surveys & Tutorials* 25 (2) (2023) 1376–1411. doi:10.1109/COMST.2023.3239220.
URL <https://doi.org/10.1109/COMST.2023.3239220>
- [3] C. Zhang, P. Patras, H. Haddadi, Deep learning in mobile and wireless networking: A survey, *IEEE Communications Surveys & Tutorials* 21 (3) (2019) 2224–2287. doi:10.1109/COMST.2019.2904897.
URL <https://doi.org/10.1109/COMST.2019.2904897>
- [4] Z. Feng, Q. Wu, X. Chen, Communication-efficient multi-service mobile traffic prediction by leveraging cross-service correlations, in: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ACM, 2024, pp. 794–805. doi:10.1145/3637528.3671730.
URL <https://doi.org/10.1145/3637528.3671730>
- [5] N. Bui, M. Cesana, S. A. Hosseini, Q. Liao, I. Malanchini, J. Widmer, A survey of anticipatory mobile networking: Context-based classification, prediction methodologies, and optimization techniques, *IEEE Communications Surveys & Tutorials* 19 (3) (2017) 1790–1821. doi:10.1109/COMST.2017.2694140.
URL <https://doi.org/10.1109/COMST.2017.2694140>
- [6] K. Alam, M. A. Habibi, M. Tammen, D. Krummacker, W. Saad, M. Di Renzo, T. Melodia, X. Costa-Pérez, M. Debbah, A. Dutta,

- H. D. Schotten, A comprehensive tutorial and survey of O-RAN: Exploring slicing-aware architecture, deployment options, use cases, and challenges, *IEEE Communications Surveys & Tutorials* 28 (2026) 1637–1678. doi:10.1109/COMST.2025.3598406.
URL <https://doi.org/10.1109/COMST.2025.3598406>
- [7] B. Brik, H. Chergui, L. Zanzi, F. Devoti, A. Ksentini, M. S. Siddiqui, X. Costa-Pérez, C. Verikoukis, Explainable AI in 6G O-RAN: A tutorial and survey on architecture, use cases, challenges, and future research, *IEEE Communications Surveys & Tutorials* 27 (5) (2025) 2826–2859. doi:10.1109/COMST.2024.3510543.
URL <https://doi.org/10.1109/COMST.2024.3510543>
- [8] C. K. Thomas, C. Chaccour, W. Saad, M. Debbah, C. S. Hong, Causal reasoning: Charting a revolutionary course for next-generation AI-native wireless networks, *IEEE Vehicular Technology Magazine* 19 (1) (2024) 16–31. doi:10.1109/MVT.2024.3359357.
URL <https://doi.org/10.1109/MVT.2024.3359357>
- [9] J. Wang, J. Tang, Z. Xu, Y. Wang, G. Xue, X. Zhang, D. Yang, Spatiotemporal modeling and prediction in cellular networks: A big data enabled deep learning approach, in: *IEEE INFOCOM 2017 – IEEE Conference on Computer Communications*, IEEE, 2017, pp. 1–9. doi:10.1109/INFOCOM.2017.8057090.
URL <https://doi.org/10.1109/INFOCOM.2017.8057090>
- [10] F. Sun, P. Wang, J. Zhao, N. Xu, J. Zeng, J. Tao, K. Song, C. Deng, J. C. S. Lui, X. Guan, Mobile data traffic prediction by exploiting time-evolving user mobility patterns, *IEEE Transactions on Mobile Computing* 21 (12) (2022) 4456–4470. doi:10.1109/TMC.2021.3079117.
URL <https://doi.org/10.1109/TMC.2021.3079117>
- [11] G. Jakobson, M. Weissman, Alarm correlation, *IEEE Network* 7 (6) (1993) 52–59. doi:10.1109/65.244794.
URL <https://doi.org/10.1109/65.244794>
- [12] A. T. Bouloutas, S. Calo, A. Finkel, Alarm correlation and fault identification in communication networks, *IEEE Transactions on Communications* 42 (2/3/4) (1994) 523–533. doi:10.1109/TCOMM.1994.577079.
URL <https://doi.org/10.1109/TCOMM.1994.577079>

- [13] A. Maatouk, N. Piovesan, F. Ayed, A. De Domenico, M. Debbah, Large language models for telecom: Forthcoming impact on the industry, *IEEE Communications Magazine* 63 (1) (2025) 62–68. doi:10.1109/MCOM.001.2300473.
URL <https://doi.org/10.1109/MCOM.001.2300473>
- [14] H. Zou, Q. Zhao, Y. Tian, L. Bariah, F. Bader, T. Lestable, M. Debbah, Telecomgpt: A framework to build telecom-specific large language models, *IEEE Transactions on Machine Learning in Communications and Networking* 3 (2025) 948–975. doi:10.1109/TMLCN.2025.3593184.
URL <https://doi.org/10.1109/TMLCN.2025.3593184>
- [15] D. Wu, X. Wang, Y. Qiao, Z. Wang, J. Jiang, S. Cui, F. Wang, NetLLM: Adapting large language models for networking, in: *Proceedings of the ACM SIGCOMM 2024 Conference*, ACM, 2024, pp. 661–678. doi:10.1145/3651890.3672268.
URL <https://doi.org/10.1145/3651890.3672268>
- [16] J. Xiao, Y. Lin, H. Qiu, R. Ma, C. Zhong, D. Xu, X. Long, C. Zhang, Q. Hao, D. Zou, Z. Yang, Y. Gao, F. Tan, Telecom-bench: How far are large language models from industrial telecommunication applications? (2026). arXiv:2605.18025.
URL <https://arxiv.org/abs/2605.18025>
- [17] C.-X. Wang, X. You, X. Gao, X. Zhu, Z. Li, C. Zhang, H. Wang, Y. Huang, Y. Chen, H. Haas, J. S. Thompson, E. G. Larsson, M. Di Renzo, W. Tong, P. Zhu, X. Shen, H. V. Poor, L. Hanzo, On the road to 6G: Visions, requirements, key technologies, and testbeds, *IEEE Communications Surveys & Tutorials* 25 (2) (2023) 905–974. doi:10.1109/COMST.2023.3249835.
URL <https://doi.org/10.1109/COMST.2023.3249835>
- [18] L. Li, M. Herrera, A. Mukherjee, G. Zheng, C. Chen, M. Dhada, H. Brice, A. Parekh, A. K. Parlikad, Predictive alarm models for improving radio access network robustness, *Expert Systems with Applications* 259 (2025) 125312. doi:10.1016/j.eswa.2024.125312.
URL <https://doi.org/10.1016/j.eswa.2024.125312>
- [19] H. Chai, S. Zhang, X. Qi, B. Qiu, Y. Li, UoMo: A universal model of mobile traffic forecasting for wireless network optimization, in: *Proceed-*

- ings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2, ACM, 2025, pp. 4308–4319. doi:10.1145/3711896.3737272.
URL <https://doi.org/10.1145/3711896.3737272>
- [20] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, M. Long, TimesNet: Temporal 2d-variation modeling for general time series analysis, in: International Conference on Learning Representations, 2023. arXiv:2210.02186.
URL https://openreview.net/forum?id=ju_Uqw3840q
 - [21] A. Zeng, M. Chen, L. Zhang, Q. Xu, Are transformers effective for time series forecasting?, Proceedings of the AAAI Conference on Artificial Intelligence 37 (9) (2023) 11121–11128. doi:10.1609/aaai.v37i9.26317.
URL <https://doi.org/10.1609/aaai.v37i9.26317>
 - [22] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long, iTransformer: Inverted transformers are effective for time series forecasting, in: International Conference on Learning Representations, 2024. arXiv:2310.06625.
URL <https://openreview.net/forum?id=JePfAI8fah>
 - [23] S. Wang, H. Wu, X. Shi, T. Hu, H. Luo, L. Ma, J. Y. Zhang, J. Zhou, TimeMixer: Decomposable multiscale mixing for time series forecasting, in: International Conference on Learning Representations, 2024. arXiv:2405.14616.
URL <https://openreview.net/forum?id=7oLshfEIC2>
 - [24] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, Y. Wang, Chronos: Learning the language of time series, Transactions on Machine Learning Research (2024). arXiv:2403.07815.
URL <https://openreview.net/forum?id=gerNCVqqtR>
 - [25] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Li, S. Pan, Q. Wen, Time-LLM: Time series forecasting by reprogramming large language models, in: International Conference

on Learning Representations, 2024. arXiv:2310.01728.
URL <https://openreview.net/forum?id=Unb5CVPtae>

- [26] F. Jia, K. Wang, Y. Zheng, D. Cao, Y. Liu, GPT4MTS: Prompt-based large language model for multimodal time-series forecasting, *Proceedings of the AAAI Conference on Artificial Intelligence* 38 (21) (2024) 23343–23351. doi:10.1609/aaai.v38i21.30383.
URL <https://doi.org/10.1609/aaai.v38i21.30383>
- [27] H. Xu, W. Chen, N. Zhao, Z. Li, J. Bu, Z. Li, Y. Liu, Y. Zhao, D. Pei, Unsupervised anomaly detection via variational auto-encoder for seasonal kpis in web applications, in: *Proceedings of the 2018 World Wide Web Conference*, ACM, 2018, pp. 187–196. doi:10.1145/3178876.3185996.
URL <https://doi.org/10.1145/3178876.3185996>
- [28] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, D. Pei, Robust anomaly detection for multivariate time series through stochastic recurrent neural network, in: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, ACM, 2019, pp. 2828–2837. doi:10.1145/3292500.3330672.
URL <https://doi.org/10.1145/3292500.3330672>
- [29] P. Fournier-Viger, G. He, M. Zhou, M. Nouioua, J. Liu, Discovering alarm correlation rules for network fault management, in: *Service-Oriented Computing – ICSOC 2020 Workshops*, Springer International Publishing, 2021, pp. 228–239. doi:10.1007/978-3-030-76352-7_24.
URL https://doi.org/10.1007/978-3-030-76352-7_24
- [30] M. Du, F. Li, G. Zheng, V. Srikumar, Deeplog: Anomaly detection and diagnosis from system logs through deep learning, in: *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, ACM, 2017, pp. 1285–1298. doi:10.1145/3133956.3134015.
URL <https://doi.org/10.1145/3133956.3134015>
- [31] G. Yu, P. Chen, Y. Li, H. Chen, X. Li, Z. Zheng, Nezha: Interpretable fine-grained root causes analysis for microservices on multi-modal observability data, in: *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ACM, 2023, pp. 553–565. doi:10.1145/361164

3.3616249.

URL <https://doi.org/10.1145/3611643.3616249>

- [32] R. Xin, P. Chen, Z. Zhao, CausalRCA: Causal inference based precise fine-grained root cause localization for microservice applications, *Journal of Systems and Software* 203 (2023) 111724. doi:10.1016/j.jss.2023.111724.
URL <https://doi.org/10.1016/j.jss.2023.111724>
- [33] C.-M. Lin, C. Chang, W.-Y. Wang, K.-D. Wang, W.-C. Peng, Root cause analysis in microservice using neural granger causal discovery, *Proceedings of the AAAI Conference on Artificial Intelligence* 38 (1) (2024) 206–213. doi:10.1609/aaai.v38i1.27772.
URL <https://doi.org/10.1609/aaai.v38i1.27772>
- [34] Z. Yao, C. Pei, W. Chen, H. Wang, L. Su, H. Jiang, Z. Xie, X. Nie, D. Pei, Chain-of-event: Interpretable root cause analysis for microservices through automatically learning weighted event causal graph, in: *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering*, ACM, 2024, pp. 50–61. doi:10.1145/3663529.3663827.
URL <https://doi.org/10.1145/3663529.3663827>
- [35] L. Pham, H. Zhang, H. Ha, F. Salim, X. Zhang, RCAEval: A benchmark for root cause analysis of microservice systems with telemetry data, in: *Companion Proceedings of the ACM on Web Conference 2025*, ACM, 2025, pp. 777–780. doi:10.1145/3701716.3715290.
URL <https://doi.org/10.1145/3701716.3715290>
- [36] M. Ma, J. Clark, S. Zhang, AIOpsLab in action: An open platform for AIOps research, in: *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, ACM, 2025, pp. 1223–1227. doi:10.1145/3696630.3728619.
URL <https://doi.org/10.1145/3696630.3728619>
- [37] Y. Ogata, Statistical models for earthquake occurrences and residual analysis for point processes, *Journal of the American Statistical Association* 83 (401) (1988) 9–27. doi:10.1080/01621459.1988.10478560.
URL <https://doi.org/10.1080/01621459.1988.10478560>

- [38] H. Xu, M. Farajtabar, H. Zha, Learning granger causality for hawkes processes, in: Proceedings of the 33rd International Conference on Machine Learning, Vol. 48 of Proceedings of Machine Learning Research, PMLR, 2016, pp. 1717–1726.
URL <https://proceedings.mlr.press/v48/xuc16.html>
- [39] J. Jiang, C. Han, W. X. Zhao, J. Wang, PDFormer: Propagation delay-aware dynamic long-range transformer for traffic flow prediction, Proceedings of the AAAI Conference on Artificial Intelligence 37 (4) (2023) 4365–4373. doi:10.1609/aaai.v37i4.25556.
URL <https://doi.org/10.1609/aaai.v37i4.25556>
- [40] H. Liu, Z. Dong, R. Jiang, J. Deng, J. Deng, Q. Chen, X. Song, Spatio-temporal adaptive embedding makes vanilla transformer SOTA for traffic forecasting, in: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, ACM, 2023, pp. 4125–4129. doi:10.1145/3583780.3615160.
URL <https://doi.org/10.1145/3583780.3615160>
- [41] X. Liu, Y. Xia, Y. Liang, J. Hu, Y. Wang, L. Bai, C. Huang, Z. Liu, B. Hooi, R. Zimmermann, LargeST: A benchmark dataset for large-scale traffic forecasting, in: Advances in Neural Information Processing Systems, Vol. 36, 2023. arXiv:2306.08259.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/ee57cd73a76bd927ffca3dda1dc3b9d4-Abstract-Datasets_and_Benchmarks.html
- [42] W. Zhang, L. Zhang, J. Han, H. Liu, Y. Fu, J. Zhou, Y. Mei, H. Xiong, Irregular traffic time series forecasting based on asynchronous spatio-temporal graph convolutional networks, in: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, ACM, 2024, pp. 4302–4313. doi:10.1145/3637528.3671665.
URL <https://doi.org/10.1145/3637528.3671665>
- [43] A. Maatouk, F. Ayed, N. Piovesan, A. De Domenico, M. Debbah, Z.-Q. Luo, Teleqna: A benchmark dataset to assess large language models telecommunications knowledge, IEEE Network 40 (2) (2026) 253–260. doi:10.1109/MNET.2025.3576035.
URL <https://doi.org/10.1109/MNET.2025.3576035>

- [44] Z.-H. Zhou, A brief introduction to weakly supervised learning, *National Science Review* 5 (1) (2018) 44–53. doi:10.1093/nsr/nwx106.
URL <https://doi.org/10.1093/nsr/nwx106>
- [45] A. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, C. Ré, Snorkel: Rapid training data creation with weak supervision, *The VLDB Journal* 29 (2–3) (2020) 709–730. doi:10.1007/s00778-019-00552-1.
URL <https://doi.org/10.1007/s00778-019-00552-1>
- [46] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: *International Conference on Learning Representations*, 2023. arXiv: 2211.14730.
URL <https://openreview.net/forum?id=Jbdc0vT0col>
- [47] A. G. Hawkes, Spectra of some self-exciting and mutually exciting point processes, *Biometrika* 58 (1) (1971) 83–90. doi:10.1093/biomet/58.1.83.
URL <https://doi.org/10.1093/biomet/58.1.83>
- [48] H. Mei, J. Eisner, The neural hawkes process: A neurally self-modulating multivariate point process, in: *Advances in Neural Information Processing Systems*, Vol. 30, 2017.
URL <https://papers.neurips.cc/paper/7252-the-neural-hawkes-process-a-neurally-self-modulating-multivariate-point-process>
- [49] Y. Li, R. Yu, C. Shahabi, Y. Liu, Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: *International Conference on Learning Representations*, 2018.
URL <https://openreview.net/forum?id=SJiHXGWAZ>
- [50] Z. Wu, S. Pan, G. Long, J. Jiang, C. Zhang, Graph wavenet for deep spatial-temporal graph modeling, in: *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, 2019, pp. 1907–1913. doi:10.24963/ijcai.2019/264.
URL <https://doi.org/10.24963/ijcai.2019/264>
- [51] A. E. Hoerl, R. W. Kennard, Ridge regression: Biased estimation for nonorthogonal problems, *Technometrics* 12 (1) (1970) 55–67. doi:10.1

080/00401706.1970.10488634.

URL <https://doi.org/10.1080/00401706.1970.10488634>

- [52] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Computation* 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.
URL <https://doi.org/10.1162/neco.1997.9.8.1735>
- [53] S. Bai, J. Z. Kolter, V. Koltun, An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, *arXiv preprint arXiv:1803.01271* (2018). arXiv:1803.01271.
URL <https://arxiv.org/abs/1803.01271>
- [54] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in Python, *Journal of Machine Learning Research* 12 (85) (2011) 2825–2830.
URL <https://jmlr.org/papers/v12/pedregosa11a.html>